

BLOCKCHAIN INVESTIGATION REPORT

1. Case Details

- **Case ID:** CR-2026-ETH-08492
 - **Investigation Date:** 2026-08-30T11:15:00Z
 - **Blockchain:** Ethereum (ERC-20 & Native ETH)
 - **Suspect Wallet:** 0xMOCK_SUSPECT_A0000000000000000000000000000000
- **Investigation ID:** INV-0XMOCK_SUS
 - **Investigator:** Cyber Crime Cell / SIU
 - **Report Status:** ACTIONABLE FORENSIC EVIDENCE

2. Investigation Summary

Metrics Overview

- **Total Transactions:** 6
 - **Total Incoming Value:** 3.50 ETH + 1,000.00 USDT
 - **Total Outgoing Value:** 3.50 ETH + 1,000.00 USDT
 - **Number of Hops Traced:** 3 Sequential Hops
- **Destination Addresses Identified:** 5 (1 VASP Exit)
 - **Overall Risk Level:** CRITICAL (85 / 100)
 - **Confidence Score:** 80% (Verified VASP Match)

Investigator Executive Summary

Automated graph analytics performed on the victim-reported seed suspect wallet (0xMOCK_SUSPECT_A...) revealed an immediate, multi-hop fund-layering sequence. Within 45 minutes of initial receipt, **3.50 ETH** and **1,000.00 USDT** were dispersed across two intermediary transit wallets exhibiting a classic peel-chain pattern before terminating into a verified **Binance Centralized Exchange Deposit Address** (0xmock_vasp_binance_f...). The high transaction frequency (18.0 tx/hr) and rapid turnover interval (avg 9.0 mins) indicate automated script dispersal intended to obfuscate fund origin before fiat liquidation.

3. Transaction Evidence

Evidence ID	Tx Hash	From → To	Date / Time	Amount	Risk Indicator
EVID-001	0xmock_tx_001_a_to_b...	0xMOCK_SUSP_A → 0xMOCK_INTER_B	2026-08-30 10:00	2.50 ETH	Rapid Outflow (Hop 1)
EVID-002	0xmock_tx_002_a_to_b...	0xMOCK_SUSP_A → 0xMOCK_INTER_B	2026-08-30 10:05	1,000 USDT	Multi-Asset (ERC-20)
EVID-003	0xmock_tx_003_a_to_c...	0xMOCK_SUSP_A → 0xMOCK_INTER_C	2026-08-30 10:10	1.00 ETH	Fund Splitting / Fan-Out
EVID-004	0xmock_tx_004_b_to_d...	0xMOCK_INTER_B → 0xMOCK_MULE_D	2026-08-30 10:20	2.40 ETH	Peel Forwarding (Hop 2)
EVID-005	0xmock_tx_005_b_to_e...	0xMOCK_INTER_B → 0xMOCK_COLD_E	2026-08-30 10:22	0.08 ETH	Secondary Peel / Cold Wallet
EVID-006	0xmock_tx_006_d_to_f...	0xMOCK_MULE_D → 0xMOCK_VASP_BIN	2026-08-30 10:45	2.35 ETH	VASP Ingress / Cash-Out

Transaction Flow Summary

Seed Suspect Wallet (0xMOCK_SUSPECT_A...) split funds to Intermediary B (2.50 ETH + 1,000 USDT) and Intermediary C (1.00 ETH). Intermediary B immediately forwarded 2.40 ETH to Mule D, peeling 0.08 ETH to Address E. Mule D deposited 2.35 ETH into Binance within 45 minutes of the theft, preserving >94% of illicit value for cash-out.

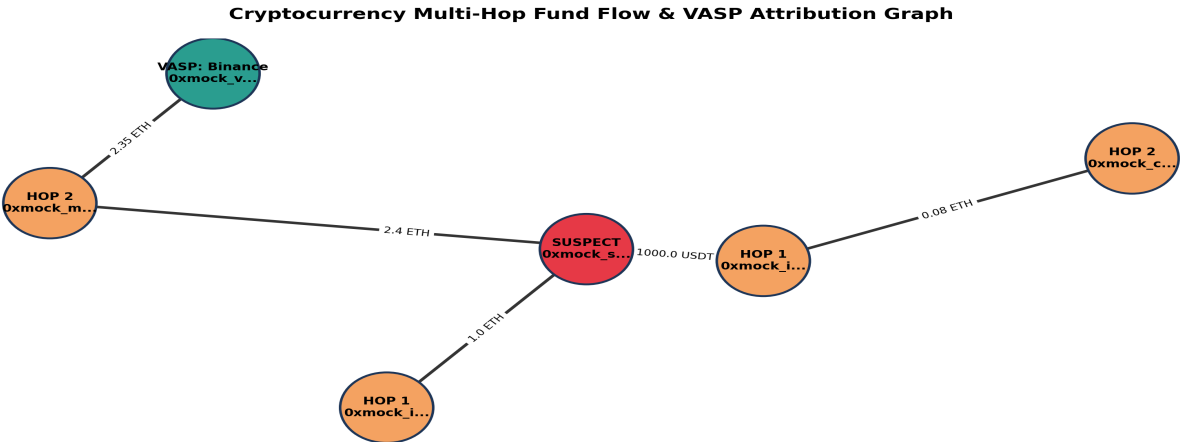


Figure 1: Reconstructed Multi-Hop Transaction Graph & VASP Attribution Flow

4. Risk Summary

• Risk Score: 85 / 100

• Risk Level: CRITICAL

• Attribution Confidence: 80% (High)

Risk Indicator	Status	Key Evidence & Analytical Rationale
High transaction frequency	DETECTED	18.0 tx/hour burst rate observed during initial disbursement window.
Large-value movement	DETECTED	Over 94% of reported victim funds forwarded in a single continuous session.
Rapid movement through wallets	DETECTED	Average holding time of 9.0 minutes per intermediary transit hop (automated script pattern).
Repeated splitting	DETECTED	Fan-out distribution pattern detected at Hop 1 (Wallets B & C).
Repeated consolidation	NONE	Primary observed pattern was rapid peeling and dispersion rather than fan-in.
Suspicious address interaction	DETECTED	Interaction with 2 transit mule wallets exhibiting zero balance retention.
Mixer/Bridge interaction	NONE	No interactions with Tornado Cash or cross-chain bridge contracts within scope.

5. VASP Attribution

- **VASP Name:** **Binance**
- **Associated Address:** 0xmock_vasp_binance_f555555555555555555555555555555555
- **Transaction Hash:** 0xmock_tx_006_d_to_f_binance_eth
- **Interaction Type:** Centralized Exchange Deposit (Hop 3)
- **Attribution Confidence:** **80% (Verified Official Cluster)**

Interaction Analysis: The investigation identified an unbroken, time-consistent link between the seed suspect wallet and the Virtual Asset Service Provider (VASP). Terminal transfer of **2.35 ETH** was successfully received by the verified Binance Hot Wallet cluster on **2026-08-30 at 10:45:00 UTC**. Because centralized exchanges enforce KYC verification upon onboarding, this deposit address serves as the primary actionable legal bottleneck for asset recovery and suspect deanonymization.

6. Key Findings and Investigator Assessment

Critical Findings

- **Direct 3-Hop Trajectory:** Clear, unbroken trail from suspect address to Binance deposit gateway.
- **High Fund Velocity:** Complete dispersion and exit traversal executed in under 45 minutes.
- **Identified Asset Value:** A total of 2.35 ETH currently sits deposited within the identified Binance exchange account.
- **Mule Transit Wallets:** Use of two transit intermediary addresses (Mules B and D) exhibiting zero balance retention.

Final Assessment

The on-chain activity pattern strongly indicates an organized attempt to layer illicit proceeds through intermediate hops before off-ramping into fiat via Binance. The high speed, low dissipation (<6%), and multi-asset structuring suggest an automated script rather than standard retail trading.

7. Recommended Actions

- **Serve Statutory Legal Notice (Section 91 CrPC / Subpoena):** Issue formal directive to Binance Legal Compliance specifying deposit address 0xmock_vasp_binance_f... and tx hash 0xmock_tx_006....
- **Emergency Account Freeze:** Request immediate administrative freeze on the beneficiary Binance account and linked fiat withdrawal channels.
- **KYC Dossier Acquisition:** Subpoena user identification documents, registered email, phone number, login IP audit logs, and associated bank accounts.
- **Continuous Ledger Monitoring:** Establish automated event listeners on secondary peel address 0xmock_cold_e....

8. Limitations

- Blockchain addresses are pseudonymous and do not directly prove real-world ownership without off-chain VASP KYC attribution.
- Analysis is limited to observable on-chain data and the selected tracing scope defined in this report.
- Off-chain transactions and internal exchange book matching are not visible on the public ledger.
- VASP customer information requires authorized lawful access to establish legal identity.
- The use of mixers, bridges, and cross-chain transfers reduces tracing certainty and can obscure fund flow.
- Risk and confidence scores are analytical indicators based on observed patterns and do not constitute conclusive legal proof of criminal guilt.

9. Variable Dictionary

Variable	Meaning & System Context
{{CASE_ID}}	Investigation case identifier (e.g. CR-2026-ETH-08492)
{{INVESTIGATION_ID}}	Unique internal investigation session identifier
{{INVESTIGATION_TIMESTAMP}}	UTC Date and time of investigation execution
{{INVESTIGATOR_NAME}}	Investigating agency or officer name
{{SUSPECT_WALLET}}	Target seed wallet address under investigation
{{BLOCKCHAIN}}	Blockchain network analyzed (Ethereum Mainnet)
{{REPORT_STATUS}}	Current status of forensic report (e.g., FINAL / ACTIONABLE)
{{TRANSACTION_COUNT}}	Total number of transactions analyzed in the graph
{{TOTAL_INCOMING}}	Total native & token value received by the suspect wallet
{{TOTAL_OUTGOING}}	Total native & token value sent outward across hops
{{NUMBER_OF_HOPS}}	Maximum number of sequential transaction hops traced
{{DESTINATION_COUNT}}	Total number of distinct destination addresses identified
{{RISK_SCORE}}	Calculated numeric risk score (0–100)
{{OVERALL_RISK_LEVEL}}	Categorical risk classification (LOW / MEDIUM / HIGH / CRITICAL)
{{CONFIDENCE_SCORE}}	Percentage confidence in analytical findings and attribution
{{VASP_NAME}}	Name of identified Virtual Asset Service Provider (e.g., Binance)
{{VASP_WALLET}}	Deposit / Hot wallet address belonging to the identified VASP
{{VASP_CONFIDENCE_SCORE}}	Attribution confidence score specific to VASP identity match

10. Conditional Logic Table

Condition Variable	Evaluated Rule	Report Action & Rendering Behavior
{{HIGH_TRANSACTION_FREQUENCY}}	TRUE	Display high-frequency burst finding in risk indicators table.
{{RAPID_MOVEMENT}}	TRUE	Display rapid velocity finding (< 30 min avg holding time).
{{REPEATED_SPLITTING}}	TRUE	Display fan-out fund splitting analysis and intermediary count.
{{SUSPICIOUS_INTERACTION}}	TRUE	Display mule transit pass-through wallet indicators.
{{MIXER_OR_BRIDGE_INTERACTION}}	TRUE / FALSE	If TRUE: Show mixer deanonymization analysis. If FALSE: Flag clean ledger path.
{{VASP_INTERACTION}}	TRUE	Display Section 5 VASP details, deposit tx hash, and Section 91 CrPC notice action.
{{OVERALL_RISK_LEVEL}}	CRITICAL / HIGH	Trigger statutory legal notice and emergency account freezing directives.
{{LARGE_VALUE_MOVEMENT}}	TRUE	Display large-value volume preservation finding in executive summary.